Assignment 1

1. Let $f(x) = \frac{2x-1}{3x-1}$. Find the composite functions $f \circ f$, $f \circ f \circ f$ and their domains.

Answer:

$$f \circ f = \frac{2 \cdot \frac{2x-1}{3x-1} - 1}{3 \cdot \frac{2x-1}{3x-1} - 1} = \frac{x-1}{3x-2}, \quad x \in (-\infty, \frac{1}{3}) \cup (\frac{1}{3}, \frac{2}{3}) \cup (\frac{2}{3}, +\infty) \tag{1}$$

$$f \circ f \circ f = \frac{\frac{2x-1}{3x-1} - 1}{3 \cdot \frac{2x-1}{3x-1} - 2} = x, \quad x \in (-\infty, \frac{1}{3}) \cup (\frac{1}{3}, \frac{2}{3}) \cup (\frac{2}{3}, +\infty) \tag{2}$$

2. Let $f(x) = \ln(x + \sqrt{x^2+1})$. Show that it is an odd function and find the inverse function of f.

Answer:

If f(x) is an odd function, it should satisfy:

$$f(x) + f(-x) = 0$$

where $f(-x) = \ln(-x + \sqrt{x^2+1})$. Since

$$f(x) + f(-x) = \ln(x + \sqrt{x^2+1}) + \ln(-x + \sqrt{x^2+1}) = \ln(x^2 + 1 - x^2) = 0 \tag{3}$$

the statement of the question is proven. Notice that

$$e^x = y + \sqrt{y^2+1}, \; e^{-x} = -y + \sqrt{y^2+1} \tag{4}$$

As results,

$$f^{-1}(x) = \frac{e^x - e^{-x}}{2}, \; x \in \mathbb{R} \tag{5}$$

3. Find the partial fraction decomposition of the following rational functions:

$$\textbf{(a)}\; \frac{3x^3 + 22x^2 + 53x + 41}{(x+2)^2(x+3)^2}, \; \textbf{(b)}\; \frac{x^4 + x^3 + x^2 - x + 1}{x(x^2+1)^2}.$$

Answer:

(a) suppose that

$$\begin{aligned} &\frac{3x^3 + 22x^2 + 53x + 41}{(x+2)^2(x+3)^2} = \frac{A_1}{x+2} + \frac{A_2}{(x+2)^2} + \frac{B_1}{x+3} + \frac{B_2}{(x+3)^2} \\ &= \frac{(A_1 + B_1)x^3 + (8A_1 + A_2 + 7B_1 + B_2)x^2 + (21A_1 + 6A_2 + 16B_1 + 4B_2)x + (18A_1 + 9A_2 + 12B_1 + 4B_2)}{(x+2)^2(x+3)^2} \end{aligned} \tag{6}$$

Every terms of the numerator of RHS should match with that of LHS, as results

$$\begin{cases} A_1 + B_1 = 3 \\ 8A_1 + A_2 + 7B_1 + B_2 = 22 \\ 21A_1 + 6A_2 + 16B_1 + 4B_2 = 53 \\ 18A_1 + 9A_2 + 12B_1 + 4B_2 = 41 \end{cases} \tag{7}$$

Applying the method Gaussian elimination:

$$
\begin{pmatrix} 1 & 0 & 1 & 0 & 3 \\ 8 & 1 & 7 & 1 & 22 \\ 21 & 6 & 16 & 4 & 53 \\ 18 & 9 & 12 & 4 & 41 \end{pmatrix}
\xrightarrow[R_4-12R_1]{\substack{R_2-7R_1\\R_3-16R_1}}
\begin{pmatrix} 1 & 0 & 1 & 0 & 3 \\ 1 & 1 & 0 & 1 & 1 \\ 5 & 6 & 0 & 4 & 5 \\ 6 & 9 & 0 & 4 & 5 \end{pmatrix}
\xrightarrow[R_4-9R_2]{R_3-6R_2}
\begin{pmatrix} 1 & 0 & 1 & 0 & 3 \\ 1 & 1 & 0 & 1 & 1 \\ -1 & 0 & 0 & -2 & -1 \\ -3 & 0 & 0 & -5 & -4 \end{pmatrix}
$$
$$
\xrightarrow[R_1+R_3]{\substack{R_4-3R_3\\R_2+R_3}}
\begin{pmatrix} 0 & 0 & 1 & -2 & 2 \\ 0 & 1 & 0 & -1 & 0 \\ -1 & 0 & 0 & -2 & -1 \\ 0 & 0 & 0 & 1 & -1 \end{pmatrix}
\xrightarrow[R_1+2R_3]{\substack{R_3+2R_4\\R_2+R_3}}
\begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 & -3 \\ 0 & 0 & 0 & 1 & -1 \end{pmatrix}
\tag{8}
$$

As results, the partial fraction decomposition of (a) is

$$
\frac{3x^3+22x^2+53x+41}{(x+2)^2(x+3)^2} = \frac{3}{x+2} - \frac{1}{(x+2)^2} - \frac{1}{(x+3)^2} \tag{9}
$$

(b) Suppose that

$$
\begin{aligned}
\frac{x^4+x^3+x^2-x+1}{x(x^2+1)^2} &= \frac{A}{x} + \frac{B_1x+C_1}{x^2+1} + \frac{B_2x+C_2}{(x^2+1)^2} \\
&= \frac{(A+B_1)x^4 + C_1x^3 + (2A+B_1+B_2)x^2 + (C_1+C_2)x + A}{x(x^2+1)^2}
\end{aligned} \tag{10}
$$

Every terms of the numerator of RHS should match with that of LHS, as results

$$
\begin{cases} A+B_1=1 \\ C_1=1 \\ 2A+B_1+B_2=1 \\ C_1+C_2=-1 \\ A=1 \end{cases}
\implies
\begin{cases} A=1 \\ B_1=0 \\ C_1=1 \\ B_2=-1 \\ C_2=-2 \end{cases} \tag{11}
$$

As results, the partial fraction decomposition of (b) is

$$
\frac{x^4+x^3+x^2-x+1}{x(x^2+1)^2} = \frac{1}{x} + \frac{1}{x^2+1} - \frac{x+2}{(x^2+1)^2} \tag{12}
$$

4. Prove the following identities:

$$
\textbf{(a)}\ \left(\frac{\sin 5\theta}{\sin\theta}\right)^2 - \left(\frac{\cos 5\theta}{\cos\theta}\right)^2 = 8\cos 2\theta(4\cos^2 2\theta - 1)\ ,
$$

$$
\textbf{(b)}\ \frac{\sin(n\theta)+\sin[(n+2)\theta]+\sin[(n+4)\theta]}{\cos(n\theta)+\cos[(n+2)\theta]+\cos[(n+4)\theta]} = \tan[(n+2)\theta]\ .
$$

Answer:

(a)

$$
LHS = \left(\frac{\sin 5\theta\cos\theta + \sin\theta\cos 5\theta}{\sin\theta\cos\theta}\right)\left(\frac{\sin 5\theta\cos\theta - \sin\theta\cos 5\theta}{\sin\theta\cos\theta}\right) = \frac{\sin 6\theta}{\sin\theta\cos\theta}\frac{\sin 4\theta}{\sin\theta\cos\theta} \tag{13}
$$

Notice that

$$
\sin 2\theta = 2\sin\theta\cos\theta,\ \sin 3\theta = -\sin^3\theta + 3\cos^2\theta\sin\theta
$$

as results,

$$
\begin{aligned}
LHS &= \frac{2(-\sin^3 2\theta + 3\cos^2 2\theta\sin 2\theta)\cdot 4(\sin 2\theta\cos 2\theta)}{\sin^2 2\theta} = 8\cos 2\theta(3\cos^2 2\theta - \sin^2 2\theta) \\
&= 8\cos 2\theta(4\cos^2 2\theta - 1) = RHS
\end{aligned} \tag{14}
$$

(b)
Notice that

$$\begin{aligned}
\sin(n\theta) &= \sin[(n+2)\theta]\cos 2\theta - \cos[(n+2)\theta]\sin 2\theta \\
\sin[(n+4)\theta] &= \sin[(n+2)\theta]\cos 2\theta + \cos[(n+2)\theta]\sin 2\theta \\
\cos(n\theta) &= \cos[(n+2)\theta]\cos 2\theta + \sin[(n+2)\theta]\sin 2\theta \\
\cos[(n+4)\theta] &= \cos[(n+2)\theta]\cos 2\theta - \sin[(n+2)\theta]\sin 2\theta
\end{aligned} \tag{15}$$

Substituting all formulas of (15) into LHS

$$\begin{aligned}
LHS &= \frac{\sin[(n+2)\theta]\cos 2\theta - \cos[(n+2)\theta]\sin 2\theta + \sin[(n+2)\theta] + \sin[(n+2)\theta]\cos 2\theta + \cos[(n+2)\theta]\sin 2\theta}{\cos[(n+2)\theta]\cos 2\theta + \sin[(n+2)\theta]\sin 2\theta + \cos[(n+2)\theta] + \cos[(n+2)\theta]\cos 2\theta - \sin[(n+2)\theta]\sin 2\theta} \\
&= \frac{\sin[(n+2)\theta](2\cos 2\theta + 1)}{\cos[(n+2)\theta](2\cos 2\theta + 1)} = \tan[(n+2)\theta] = RHS
\end{aligned} \tag{16}$$

5. Solve the following equations:

(a) $2\cos\theta\cos 2\theta + \sin 2\theta = 2\left(3\cos^3\theta - \cos\theta\right)$, **(b)** $\cosh 4x + 4\cosh 2x - 125 = 0$

Answer:

(a)

$$2\cos\theta\left(2\cos^2\theta - 1\right) + \sin 2\theta = 6\cos^3\theta - 2\cos\theta \tag{17}$$

$$\sin 2\theta = 2\cos^3\theta \longrightarrow \sin\theta\cos\theta = \cos^3\theta \tag{18}$$

If $\cos\theta = 0$, that is $\theta = \frac{\pi}{2} + k\pi,\ k \in \mathbb{Z}$;
if $\cos\theta \neq 0$,

$$\sin\theta = \cos^2\theta = 1 - \sin^2\theta \longrightarrow \sin^2 + \sin\theta - 1 = 0 \tag{19}$$

$$\sin\theta = \frac{-1-\sqrt{5}}{2} \quad or \quad \frac{-1+\sqrt{5}}{2}$$

since $\sin\theta \in [-1, 1]$

$$\sin\theta = \frac{-1+\sqrt{5}}{2} \tag{20}$$

As results, the solutions of this equation are:

$$\theta = \frac{\pi}{2} + k\pi \quad or \quad \arcsin\frac{-1+\sqrt{5}}{2} + 2k\pi \quad or \quad \arcsin\frac{1-\sqrt{5}}{2} + (2k+1)\pi,\ k \in \mathbb{Z} \tag{21}$$

(b)

$$LHS = 2\cosh^2 2x + 4\cosh 2x - 126 = 0 \longrightarrow (\cosh 2x - 7)(\cosh 2x + 9) = 0 \tag{22}$$

$$\cosh 2x = 7 \quad or \quad -9$$

Since $\cosh x \in [1, +\infty)$, $\cosh 2x = 7$ As results, the solutions of the equation are

$$x = \pm\frac{1}{2}\cosh^{-1}(7) \tag{23}$$

6. Prove the following equations:

(a) $\sin^{-1}\frac{x-1}{x+1} = 2\tan^{-1}\sqrt{x} - \frac{\pi}{2}$, **(b)** $\tanh(\ln x) = \frac{x^2-1}{x^2+1}$

Answer:

(a)
Suppose that

$$\tan^{-1}\sqrt{x} = \theta$$

The original equation changes to

$$\sin^{-1}\left(\frac{\tan^2\theta - 1}{\tan^2\theta + 1}\right) = 2\theta - \frac{\pi}{2} \tag{24}$$

What we need to prove is now

$$\frac{\tan^2\theta - 1}{\tan^2\theta + 1} = -\cos 2\theta \tag{25}$$

Notice that

$$RHS = \cos^2\theta - \sin^2\theta = \frac{\cos^2\theta - \sin^2\theta}{\cos^2\theta + \sin^2\theta} = -\frac{\tan^2\theta - 1}{\tan^2\theta + 1} \tag{26}$$

The statement of the question is proven!

(b)
Notice that

$$\tanh\theta = \frac{\sinh\theta}{\cosh\theta} = \frac{e^x - e^{-x}}{e^x + e^{-x}} \tag{27}$$

substitute ln x into equation(27)

$$\tanh(\ln x) = \frac{x - \frac{1}{x}}{x + \frac{1}{x}} = \frac{x^2 - 1}{x^2 + 1} = RHS \tag{28}$$

7. Without calculator, evaluate **(a)** $\cos[\pi \sinh(\ln 2)]$, **(b)** $\cosh^{-1}[\coth(\ln 3)]$.

Answer:

(a)

$$\cos[\pi \sinh(\ln 2)] = \cos[\pi\frac{e^{\ln 2} - e^{-\ln 2}}{2}] = \cos(\frac{3\pi}{4}) = -\frac{\sqrt{2}}{2} \tag{29}$$

(b)

$$\begin{aligned} \cosh^{-1}[\coth(\ln 3)] &= \cosh^{-1}[\frac{e^{\ln 3} + e^{-\ln 3}}{e^{\ln 3} - e^{-\ln 3}}] = \cosh^{-1}\frac{5}{4} \\ &= \ln\left(\frac{5}{4} + \sqrt{\left(\frac{5}{4}\right)^2 - 1}\right) = \ln 2 \end{aligned} \tag{30}$$

8. Evaluate the following limits:

$$\textbf{(a)}\lim_{x\to 1}(x-1)^2\left[4 + \cos^2\left(\frac{2\pi}{x-1}\right)\right], \quad \textbf{(b)}\lim_{x\to\infty}(\sqrt{x^2+x} - \sqrt{x^2-x})$$

Answer:

(a)
Notice that

$$4(x-1)^2 < (x-1)^2\left[4 + \cos^2\left(\frac{2\pi}{x-1}\right)\right] < 5(x-1)^2 \tag{31}$$

and

$$\lim_{x\to 1} 5(x-1)^2 = \lim_{x\to 1} 4(x-1)^2 = 0 \tag{32}$$

According to the squeeze theorem,

$$\lim_{x\to 1}(x-1)^2\left[4+\cos^2\left(\frac{2\pi}{x-1}\right)\right] = 0 \tag{33}$$

(b)
Notice that

$$\sqrt{x^2+x}-\sqrt{x^2-x} = \frac{x^2+x-x^2+x}{\sqrt{x^2+x}+\sqrt{x^2-x}} = \frac{2}{\sqrt{1+\frac{1}{x}}+\sqrt{1-\frac{1}{x}}} \tag{34}$$

As results,

$$\lim_{x\to\infty}(\sqrt{x^2+x}-\sqrt{x^2-x}) = \lim_{x\to\infty}\frac{2}{\sqrt{1+\frac{1}{x}}+\sqrt{1-\frac{1}{x}}} = 1 \tag{35}$$

9. Evaluate the following limits:

$$\textbf{(a)}\lim_{x\to\infty}\left(\frac{\cos x}{\cos 2x}\right)^{1/x^2}, \quad \textbf{(b)}\lim_{x\to 0}\left(\frac{1+x2^x}{1+x3^x}\right)^{1/x^2}$$

Answer:

(a)

$$\lim_{x\to 0}\left(\frac{\cos x}{\cos 2x}\right)^{1/x^2} = \exp\left[\lim_{x\to 0}\frac{1}{x^2}\ln\left(\frac{\cos x}{\cos 2x}\right)\right] \tag{36}$$

So now we need to calculate

$$\lim_{x\to 0}\frac{1}{x^2}\ln\left(\frac{\cos x}{\cos 2x}\right)$$

Notice that

$$\lim_{x\to 0}\ln\left(\frac{\cos x}{\cos 2x}\right) = 0, \quad \lim_{x\to 0} x^2 = 0 \tag{37}$$

using L'hospital's role:

$$\begin{aligned}\lim_{x\to 0}\frac{1}{x^2}\ln\left(\frac{\cos x}{\cos 2x}\right) &= \lim_{x\to 0}\frac{1}{2x}\frac{\cos 2x}{\cos x}\frac{-\sin x\cos 2x+2\sin 2x\cos x}{\cos^2 2x} \\ &= \lim_{x\to 0}\frac{\sin x+\sin 2x\cos x}{2x\cos x\cos 2x}\end{aligned} \tag{38}$$

using L'hospital's role again:

$$\lim_{x\to 0}\frac{\sin x+\sin 2x\cos x}{2x\cos x\cos 2x} = \lim_{x\to 0}\frac{\cos x+2\cos 2x\cos x-2\sin 2x\sin x}{2\cos x\cos 2x-2x\sin x\cos x-4x\cos x\sin 2x} = \frac{3}{2} \tag{39}$$

As results,

$$\lim_{x\to 0}\left(\frac{\cos x}{\cos 2x}\right)^{1/x^2} = e^{3/2} \tag{40}$$

(b)

$$\lim_{x\to 0}\left(\frac{1+x2^x}{1+x3^x}\right)^{1/x^2} = \exp\left[\lim_{x\to 0}\frac{1}{x^2}\ln\left(\frac{1+x2^x}{1+x3^x}\right)\right] \tag{41}$$

using L'hosptal's role,

$$\begin{aligned}\lim_{x\to 0}\frac{1}{x^2}\ln\left(\frac{1+x2^x}{1+x3^x}\right) &= \lim_{x\to 0}\frac{2^x-3^x+x(\ln 2\; 2^x-\ln 3\; 3^x)+x^2 2^x 3^x(\ln 2-\ln 3)}{2x(1+x2^x)(1+x3^x)}\\ &= \lim_{x\to 0}\frac{2(\ln 2\; 2^x-\ln 3\; 3^x)+x(\ln^2 2\; 2^x-\ln^2 3\; 3^x)+\cdots}{2(1+x2^x)(1+x3^x)+2x\cdots}\\ &= \ln 2-\ln 3\end{aligned} \tag{42}$$

The ellipses represent terms which are timed by x and will vanish when $x \to 0$. As results,

$$\lim_{x\to 0}\left(\frac{1+x2^x}{1+x3^x}\right)^{1/x^2} = e^{\ln 2-\ln 3} = \frac{2}{3} \tag{43}$$

10. Find the location x_0 of any discontinuities in the following functions and classify them as removable or non-removable.In the former case, specify the redefined value $f(x_0)$ required to remove the discontinuity.

$$\textbf{(a)}\,\frac{x^3-3x^2+3x-2}{x^2+3x-10}, \qquad \textbf{(b)}\,\frac{x+3}{4-\sqrt{x^2+7}}$$

Answer:

(a)

$$\frac{x^3-3x^2+3x-2}{x^2+3x-10} = x-6+\frac{21x-62}{(x-2)(x+5)} \tag{44}$$

when $x = 2$ and $x = -5$, the function is discontinued and it is non-removable.

(b)

$$\frac{x+3}{4-\sqrt{x^2+7}} = \frac{(x+3)(4+\sqrt{x^2+7})}{(3-x)(3+x)} \tag{45}$$

When $x = \pm 3$ the function is discontinued. $x = 3$ is non-removable whereas $x = -3$ is removable and $f(-3) = \frac{4}{3}$.